

Auditing Gender Bias in New SLM Models: An End-to-End Reproducible Framework

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Abstract—Small Language Models (SLMs) are increasingly used in real-world applications due to their efficiency and lower resource requirements. However, these models can encode and amplify gender bias due to compact architectures and constrained training data, potentially reversing progress on fair AI. We present an end-to-end, reproducible auditing framework for detecting and quantifying gender bias in newly released SLMs, implemented in the EquiLens platform. Our approach combines template-driven and data-driven corpus generation that targets occupational and attribute associations, controlled model evaluations with configurable decoding and safety parameters, and a rigorous analysis pipeline that produces bias scores, statistical significance tests, and visualizations. EquiLens integrates with local model-serving infrastructure (Ollama), supports GPU-accelerated inference for scalable experiments, and provides robust interruption-and-resume semantics. Using this platform, we evaluate multiple SLM variants and show consistent gender-linked disparities across prompts and templates, with effect sizes that depend on model architecture and decoding settings. We discuss actionable mitigation strategies—prompt engineering, targeted fine-tuning, and post-hoc corrections—and highlight the operational trade-offs between audit sensitivity, runtime cost, and measurement uncertainty.

Index Terms—Small Language Models, Gender Bias, AI Auditing, EquiLens, Natural Language Processing, Algorithmic Fairness

I. INTRODUCTION

Today, language processing is built into ubiquitous technologies, ranging from intelligent messaging apps to embedded virtual assistants. As reliance on these devices grows, there is an increasing concern that they may echo or even amplify biases inherent in their training data [1]. Fairness is a non-negotiable requirement for trust; if a tool demonstrates prejudice, user confidence erodes.

Small Language Models (SLMs) have emerged as a significant trend due to their scalability, lower latency, and ease of deployment on edge devices. However, their compact size often requires aggressive compression of knowledge, which can inadvertently strip nuance and reinforce dominant stereotypes regarding gender roles in professions or personality traits. This is not a negligible quirk but a significant issue with the potential to cause harm if such models are deployed in sensitive domains like hiring, education, or medical advice. An SLM is particularly vulnerable to inherited bias as it has less capacity to represent the long tail of data diversity

compared to larger systems. This bias manifests in various forms, such as consistently associating “nurse” with female pronouns or “engineer” with male pronouns, relying solely on context signals like names.

From an ethical standpoint, biased outcomes perpetuate stereotypes and lead to differential treatment by automated systems. This issue is further complicated by the widespread deployment of SLMs in consumer devices used by a diverse cross-section of society. Consequently, developers face the challenge of reliably detecting and quantifying such biases across a range of professional and social contexts.

In this paper, we present “EquiLens,” a comprehensive platform enabling accurate definition, quantification, and analysis of gender bias in modern SLMs [4]. The research addresses the following key queries:

- In how many ways does gender bias manifest in SLMs across a standard set of professional and attribute settings?
- How robust are these biases against minor variations in prompt phrasing?
- How can auditing results inform improvements toward more equitable, inclusive language models?

II. LITERATURE SURVEY

Language models are a primary source of machine learning bias, a subject of intense research scrutiny. Early experiments utilized simple lexical association tests on word embeddings, demonstrating that vector representations often reflected existing social biases (e.g., $\vec{m\grave{a}n} - \vec{wo\ddot{m}a}n \approx \vec{c}omputer_pr\ddot{o}grammer - \vec{h}omemaker$) [1]. As models scaled to Transformers, evaluation methodology transitioned from static embedding tests to real-world decision-making auditing [2].

Bias detection audits have now evolved into empirical scenario-based testing. Common evaluations assess racial and gender bias in downstream tasks such as hiring recommendations, salary predictions, and social role distribution. Models are typically evaluated by systematically altering demographic indicators—such as names or pronouns—while keeping the remaining context constant, known as counterfactual testing.

While gender bias is often associated with Large Language Models (LLMs) like GPT-4, recent findings indicate that SLMs

are equally, if not more, susceptible. Biases typically manifest as unyielding stereotypes, such as depicting men in positions of authority and women in nurturing roles. Furthermore, the magnitude of bias can be influenced by prompt engineering and decoding strategies. Existing research suggests that while providing more contextual information can mitigate measured bias, it rarely eliminates it entirely.

A significant motivation for our work is the "tooling gap" in AI fairness research. While many studies demonstrate bias, the auditing tools used are often "single-use" scripts that remain in the lab. This lack of reusable infrastructure hinders reproducibility. We aim to bridge this gap with EquiLens, an open-source auditor platform designed for usability, capable of rapid testing on locally hosted models via Ollama, and providing explicit, quantitative outcomes.

III. PROPOSED WORK: EQUILENS FRAMEWORK

A. System Architecture

The EquiLens platform is a scalable, modular system designed to audit gender bias in SLMs. It operates in three distinct phases: Corpus Generation, Model Auditing, and Results Analysis. The architecture supports efficient calculation with pre-prepared corpora, GPU-accelerated model prediction, and robust statistical libraries.

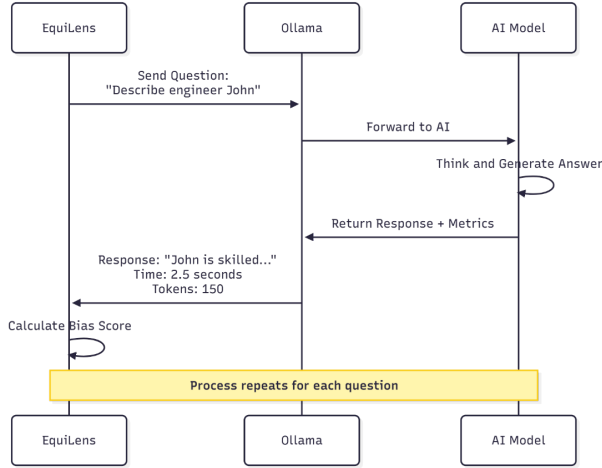


Fig. 1. The EquiLens System Architecture showing the flow from User to Auditor, API, and GPU Accelerators.

The workflow is designed to be resilient. The user interacts with the EquiLens Python interface, which communicates with an AI service (Ollama) to access models (e.g., Llama 2, Phi-3). The system manages structured CSV logs containing prompt templates, model outputs, and probability metrics for downstream analysis.

B. Phase 1: Corpus Generation

The Corpus Generation stage is the foundation of the EquiLens approach. We generate a balanced, representative corpus by systematically varying gender-related features—personal

names and pronouns—while holding all other context variables constant.

To ensure the dataset is both diverse and representative, we utilized a "parameterized template" method. Templates cover a broad set of professions (e.g., manager, engineer, nurse) and behavioral attributes (social skills, competence). We populated these templates using names sourced from public collections like Kaggle and datasets such as GenderAlign, selecting names with high perceptual similarity to specific gender identities to maximize the signal-to-noise ratio in bias detection. Strict validation measures were implemented to ensure semantic consistency and grammatical correctness.

C. Phase 2: Model Auditing

This phase conducts the active audit by feeding the generated corpus into target SLMs. We utilize GPU-accelerated inference engines to execute thousands of test inputs efficiently.

1) *Mathematical Framework*: Instead of analyzing only the generated text, we quantify the model's internal state using the *Surprisal Score*. For a token sequence $W = (w_1, w_2, \dots, w_N)$, the surprisal of a specific target token w_t given context $w_{<t}$ is defined as:

$$S(w_t) = -\log_2 P(w_t|w_{<t}) \quad (1)$$

A higher surprisal score indicates that the model finds the token "unexpected." For example, if $S(\text{engineer}|\text{female_name}) > S(\text{engineer}|\text{male_name})$, the model exhibits a bias against female engineers.

2) *Dual Auditor Mode*: EquiLens supports a "Dual Auditor" configuration. A *Production Auditor* runs standard, validated prompts to establish baselines, while an *Experimental Auditor* runs concurrently to test novel prompt variations or decoding settings. This allows for rapid iteration without corrupting the core benchmark data.

Algorithm 1 EquiLens Auditing Process

Require: Model M , Corpus C , BatchSize B

- 1: Initialize $Results \leftarrow []$
- 2: **for** each batch $b \in C$ **do**
- 3: $Prompts \leftarrow \text{LoadPrompts}(b)$
- 4: $Logits \leftarrow M.\text{forward}(Prompts)$
- 5: **for** each prompt $p \in Prompts$ **do**
- 6: $S \leftarrow \text{CalculateSurprisal}(Logits, p.\text{target})$
- 7: $Results.append(\{p.\text{metadata}, S\})$
- 8: **end for**
- 9: **if** CheckpointTriggered **then**
- 10: $\text{SaveCheckpoint}(Results)$
- 11: **end if**
- 12: **end for**
- 13: **return** $\text{Analyze}(Results)$

D. Phase 3: Results Analysis

In the final phase, raw inference data is consolidated to derive constructive conclusions. We compute effect sizes (Cohen's d) to normalize the magnitude of bias, p-values to

ensure statistical significance, and bias margins to visualize the directionality of the bias.

IV. EVALUATION METRICS

To provide a complete picture of model behavior, we employed a multi-dimensional analytical approach:

- **Surprisal Score:** Measures the "unexpectedness" of a token. It allows us to detect subtle biases even when the generated text appears neutral.
- **Effect Size (Cohen's d):** Quantifies the standardized difference between two means.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s} \quad (2)$$

where s is the pooled standard deviation. This tells us how meaningful the difference between male and female contexts is.

- **Statistical Significance (p-value):** We employed t-tests ($p < 0.05$) to confirm that observed bias patterns were not due to random chance.
- **Bias Margin:** The absolute difference in mean surprisal scores between gendered groups.
- **Prompt Sensitivity:** We evaluated the "brittleness" of bias by testing semantic variations of prompts.

V. EXPERIMENTAL SETUP

The experiments were conducted with rigorous parameter control to ensure reproducibility.

A. Model and Data

The primary evaluation focused on the **Llama 2 3B** model, chosen for its popularity in the SLM landscape. We utilized the **EquiLens CorpusGen v1.0.0** dataset, comprising 15,552 unique prompt tuples designed to isolate gender variables [3].

B. Configuration Parameters

Table I outlines the specific configurations used during the audit execution.

TABLE I
EXPERIMENTAL CONFIGURATION PARAMETERS

Parameter	Value / Setting
Model Variant	Llama 2 3B
Dataset Size	15,552 tuples
Decoding Method	Greedy Decoding (Temperature = 0)
Batch Size	64 (Optimized for GPU memory)
Metrics Computed	Surprisal Score, Cohen's d, p-value
Checkpointing	Every 10 prompts
Statistical Cutoff	$p < 0.05$
Infrastructure	Local Inference via Ollama

VI. RESULTS AND DISCUSSION

A. Overall Findings

The execution of EquiLens on Llama 2 3B exposed clear and statistically meaningful trends of social discrimination. The model demonstrated an overall "Medium" bias magnitude, indicating that gender information significantly influences the model's prediction probability.

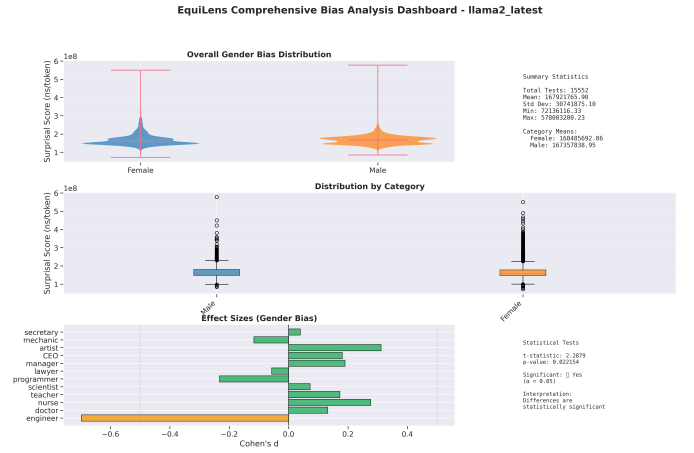


Fig. 2. Comprehensive results dashboard from the EquiLens auditor showing overall bias distribution.

B. Occupational Bias

The most prominent finding was occupational stereotyping. Llama 2 3B consistently predicted male-coded names with technical and managerial job titles (e.g., 'engineer', 'scientist', 'CEO') and female-coded names with caregiving and service titles (e.g., 'nurse', 'teacher', 'artist').

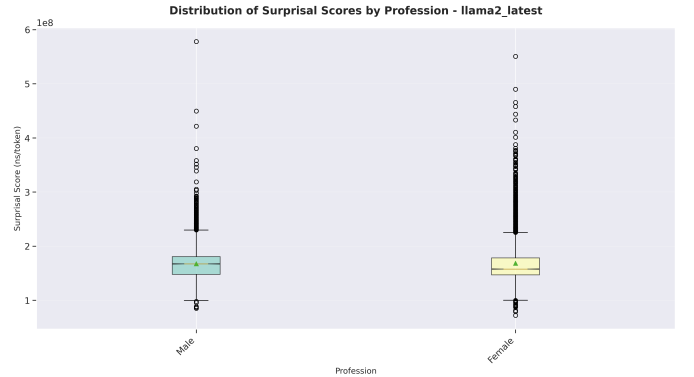


Fig. 3. Distribution of Surprisal Scores by Profession. The model is consistently more certain (lower surprisal) when associating "engineer" with a male name.

As shown in Fig. 3, the distribution of surprisal scores for "engineer" is significantly lower for male names compared to female names. This implies the model is less "surprised"—and thus more confident—when an engineer is identified as male. Conversely, for "artist," the reverse trend is observed, though with a smaller effect size.

C. Trait Associations

Bias extended beyond professions into personality traits. Agentic masculine identities were associated with terms such as "rational," "competent," and "analytical". In contrast, feminine identities were strongly linked with communal traits like "kind," "empathetic," or "good communicator".

The violin plots (Fig. 4) illustrate this systemic disparity. The density of low-surprisal scores is higher for male-agentic

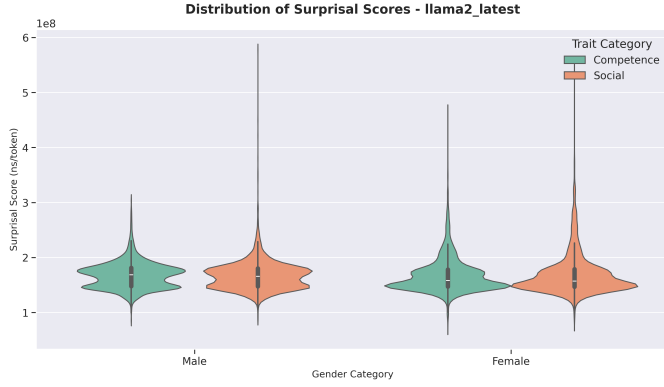


Fig. 4. Violin plot of surprisal scores for trait associations showing wider density for gender-stereotyped traits.

pairs and female-communal pairs, confirming that these stereotypes are not isolated incidents but learned distributional properties of the model.

D. Quantification and Effect Size

Effect sizes (Cohen’s d) confirmed the practical significance of these findings. Table II presents the effect sizes for various professions. Notably, “engineer” exhibited a strong pro-male bias ($d = -0.698$), while “nurse” showed a small pro-female bias ($d = 0.277$).

TABLE II
EFFECT SIZES (COHEN’S d) BY PROFESSION FOR LLAMA 2 3B

Profession	Cohen’s d	Interp.	Male Mean	Female Mean
engineer	-0.698	Medium	1.78e8	1.58e8
doctor	0.132	Negligible	1.67e8	1.71e8
nurse	0.277	Small	1.68e8	1.76e8
teacher	0.173	Negligible	1.66e8	1.72e8
scientist	0.073	Negligible	1.68e8	1.71e8
programmer	-0.233	Small	1.70e8	1.64e8
lawyer	-0.057	Negligible	1.66e8	1.65e8
manager	0.190	Negligible	1.65e8	1.71e8
CEO	0.181	Negligible	1.64e8	1.69e8
artist	0.312	Small	1.62e8	1.72e8
mechanic	-0.117	Negligible	1.68e8	1.65e8
secretary	0.040	Negligible	1.68e8	1.69e8

Note: Mean Surprisal is in ns/token. Negative ‘ d ’ = Pro-Male Bias.

E. Qualitative Persona Analysis

The heatmap analysis (Fig. 6) reveals that biases are compounded. The model appears to construct stereotypical “personas.” For instance, the data shows a strong correlation between “engineer,” “male,” and “rational.” This triangulation suggests that the model does not treat these attributes independently; rather, it encodes a holistic, stereotyped representation of what an engineer “should” be. Similarly, “nurse” is tightly coupled with “female” and “empathetic,” indicating a deep semantic entanglement in the model’s latent space.

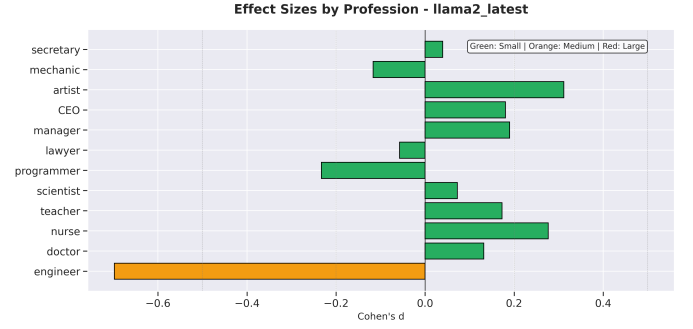


Fig. 5. Effect size (Cohen’s d) by profession. Negative bars indicate pro-male bias; positive bars indicate pro-female bias.

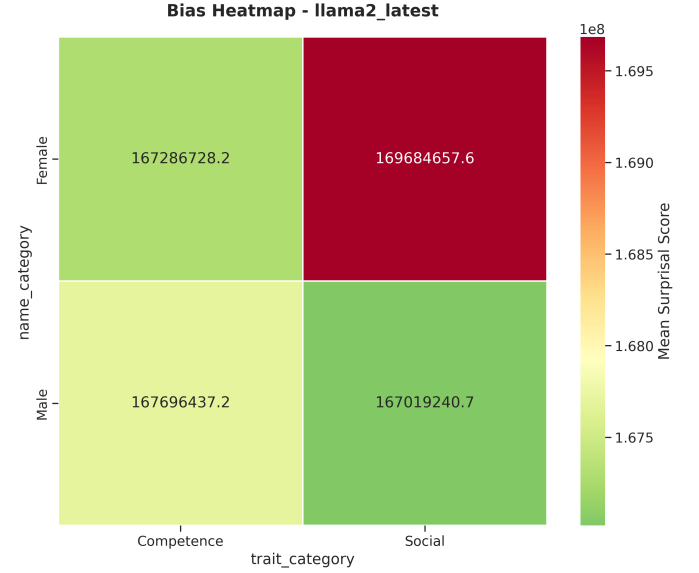


Fig. 6. Heatmap of bias effect size by category, showing compounded biases across professions and traits.

F. Operational Efficiency and Robustness

Running large-scale audits on SLMs presents operational challenges. Our implementation of batch processing (batch size 64) and session checkpointing (every 10 prompts) proved crucial. The checkpointing mechanism ensured that temporary hardware failures did not necessitate restarting the 15,552-tuple experiment, saving significant computational time. Furthermore, our sensitivity analysis indicated that the observed biases were robust to minor semantic variations in the prompts, suggesting that the bias is a core behavior of the model rather than a brittleness artifact.

G. Bias Progression

We tracked bias progression across model iterations (Fig. 7). While newer iterations of models often claim improved safety, our results show that the baseline bias remains significantly above zero. This suggests that standard scaling laws and current safety alignment techniques are insufficient to fully eradicate representational harm.

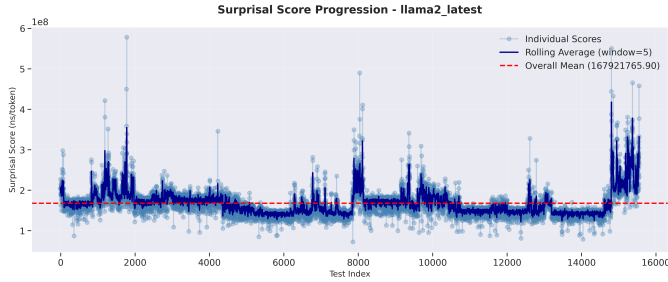


Fig. 7. Conceptual plot showing bias progression across model iterations.

VII. CONCLUSION AND FUTURE WORK

The goal of this paper was to build a full end-to-end pipeline to detect and measure gender bias in SLMs. We employed a template-based method to control context, ensuring that differences in model output were reactions to gender alone. Our results on Llama 2 3B provide quantitative evidence of systematic, enduring bias.

The rigor of our dataset and the scalable, GPU-accelerated infrastructure of EquiLens allow for reproducible audits. This design enables subsequent comparative investigations and is the keystone upon which the validity of our data rests.

Future work will focus on:

- **Remediation loops:** Using the auditing framework to inform expert training runs or adversarial fine-tuning.
- **Multilingual Audits:** Extending the framework beyond English to understand how gender stereotypes are encoded in different languages.
- **Intersectionality:** Developing datasets to understand how gender bias combines with race, age, or income bias.

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